Chapter 1: Fundamentals of Mathematics

1.1 Introduction

Mathematics is the study of numbers, quantities, shapes, and patterns. It provides the language and tools needed to describe the physical world precisely and rigorously. From counting apples to launching satellites, mathematics underpins every scientific discipline.

The **fundamental theorem of calculus** connects differentiation and integration. It states that if *f* is continuous on [a, b] then the function F defined by F(x) = integral from a to x of f(t) dt is differentiable and F'(x) = f(x).

1.1.1 Key Definitions

Derivative: The derivative of a function f at a point x is defined as the limit of (f(x+h) - f(x)) / h as h approaches zero, provided the limit exists.

Integral: The definite integral of f over [a, b] represents the net signed area between the curve y = f(x) and the x-axis over that interval.

1.2 Algebraic Foundations

Core algebraic properties:

- Commutative property: a + b = b + a
- Associative property: (a + b) + c = a + (b + c)
- Distributive property: a * (b + c) = a*b + a*c
- Identity element: a + 0 = a
- Inverse element: a + (-a) = 0

Steps to solve a quadratic equation ax^2 + bx + c = 0:

1. Identify the coefficients a, b, and c.
2. Compute the discriminant D = b^2 - 4ac.
3. If D >= 0, apply the quadratic formula: x = (-b ± sqrt(D)) / (2a).
4. If D < 0, the roots are complex conjugates.

1.3 Important Equations

Quadratic formula:

```
x = (-b ± sqrt(b^2 - 4ac)) / (2a) ... (Eq. 1.1)
```

Euler's identity:

```
e^(i*pi) + 1 = 0 ... (Eq. 1.2)
```

Pythagorean theorem:

```
a^2 + b^2 = c^2 ... (Eq. 1.3)
```

Area of a circle:

```
A = pi * r^2 ... (Eq. 1.4)
```

1.4 Trigonometric Values

The following table shows common trigonometric function values.

Angle (deg)	sin(θ)	cos(θ)	tan(θ)
0°	0	1	0
30°	1/2	√3/2	1/√3
45°	√2/2	√2/2	1
60°	√3/2	1/2	√3
90°	1	0	undefined

1.5 Numerical Methods

The Newton-Raphson method iteratively approximates roots of f(x) = 0 using:

```
def newton_raphson(f, df, x0, tol=1e-6, max_iter=100):
    x = x0
    for i in range(max_iter):
        fx = f(x)
        if abs(fx) < tol:
            return x
        x = x - fx / df(x)
    raise ValueError('Did not converge')
```

TIP: When implementing Newton-Raphson, always check that df(x) != 0 to avoid division by zero. Add a guard clause before the division step.

1.6 Calculus Review

The derivative of common functions is summarized below. These rules are applied extensively in optimization problems.

Power rule:

```
d/dx [x^n] = n * x^(n-1) ... (Eq. 1.5)
```

Chain rule:

```
d/dx [f(g(x))] = f'(g(x)) * g'(x) ... (Eq. 1.6)
```

Product rule:

```
d/dx [u*v] = u'*v + u*v' ... (Eq. 1.7)
```

1.7 Summary

In this chapter we reviewed the fundamental building blocks of mathematics: algebraic properties, key definitions from calculus, important equations, trigonometric identities, and numerical methods. These concepts form the backbone of all advanced study in science and engineering.